Respiratory Failure & ARDS: An Approach to a Case of Cyanosis

Dr. Essam Kotb, Internal Medicine | Source: `14- RF & ARDS updated.pdf` (34 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The two case answers are **reasoned by me; the key is not on the slides**.

🔗 **Bridge from earlier lectures:**

- **COPD / asthma:** AECOPD and acute severe asthma are the classic **type 2** failures; the COPD **O_2 target is 88–92%**.
- **Sleep apnoea / OHS:** OHS is **chronic type 2** failure, "often diagnosed in acute respiratory failure"; ↑ HCO_3^- shows chronicity.
- **Pneumonia:** the commonest **direct** cause of ARDS and of type 1 failure.
- **Pulmonary vascular:** PE = V/Q mismatch; Eisenmenger = intracardiac shunt.
- **Restrictive lung disease:** IPF = diffusion defect; neuromuscular/chest-wall restriction = pump failure.
- **Psychiatry (substances/psychotropics): opioid and benzodiazepine overdose → CNS depression → type 2 failure**; cocaine is a systemic cause of ARDS.
- **Upcoming (Sepsis):** sepsis is the **most common cause of ARDS**.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ **Type 1 vs Type 2**; acute vs chronic	ABG classification
Knowledge	Mechanisms of type 1 and 2 failure; **diffuse alveolar damage** in ARDS	Shunt; hyaline membranes
Knowledge	**Direct and indirect** causes of ARDS	Pancreatitis, sepsis, TRALI
Knowledge	⭐ Work-up; **Global Definition (2023) / Berlin criteria**	P/F thresholds
Knowledge	**Central vs peripheral cyanosis**	DDx
Skill	Analyse ABGs; **oxygenation ratios** (P/F, S/F)	Severity
Skill	Systematic work-up: bedside, labs, CXR, CT, **POCUS**	
Skill	O_2 delivery; **ABCDE**	
Skill	⭐ **Lung-protective ventilation**: low tidal volume, low plateau pressure, PEEP	Lecture case 2
Values	Early recognition and resuscitation; ARDSNet protocols; ethics in critical care	

Main subtopics:

1. **Type 1 respiratory failure**
2. **Type 2 respiratory failure**
3. **ARDS**

4. **Cyanosis**: differential diagnosis and types

2. Main Content (Lecture Order)

2.1 Definitions and Classification (Slides 5–6) ⭐⭐

- **Respiratory failure:** inability of the respiratory system to maintain **adequate gas exchange** (normal O_2 and CO_2); acute or chronic.
  - ⭐ **Hypoxaemic: PaO_2 < 60 mmHg**
  - ⭐ **Hypercapnic: $PaCO_2$ > 45 mmHg** (normal 35–45)
- **Respiratory arrest:** complete cessation of breathing **with an intact pulse**.

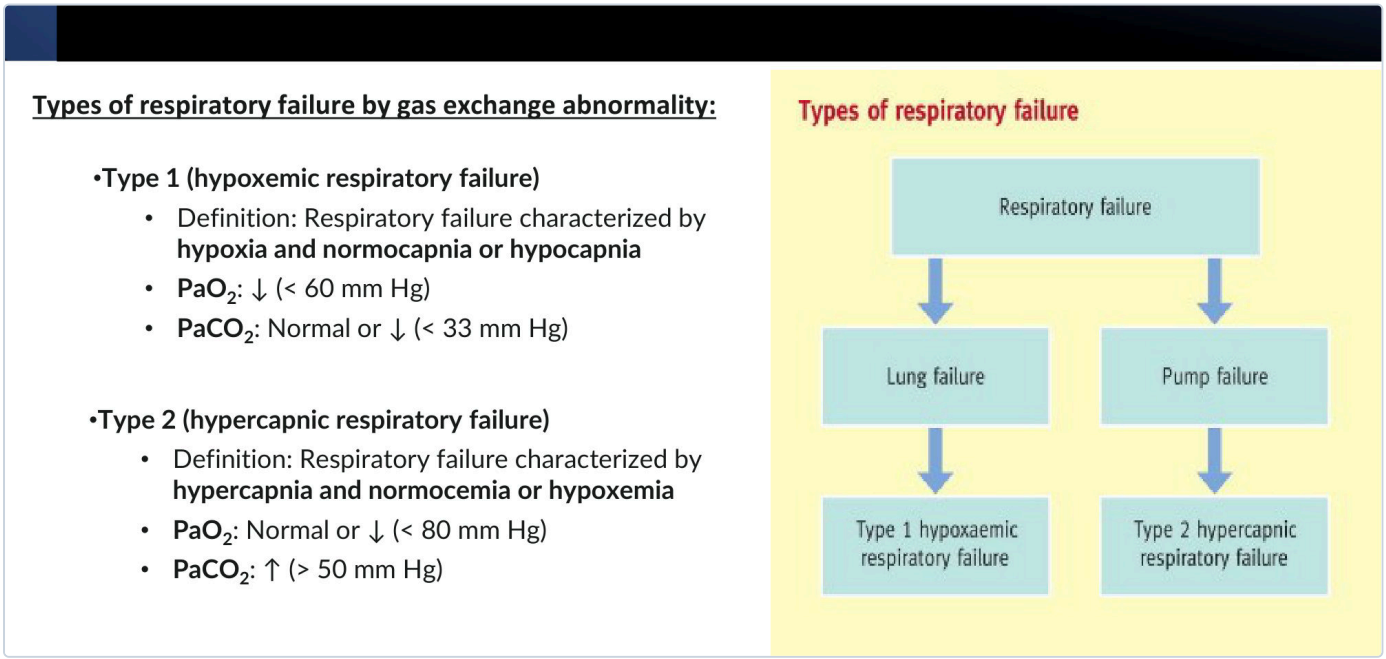
By duration	
Acute	Develops over **minutes to hours** (acute illness or insult)
Chronic	Long-standing, from chronic illness: **COPD, ILD, OHS**

By gas exchange	**Type 1 (hypoxaemic)**	**Type 2 (hypercapnic)**
Definition	Hypoxia + **normocapnia or hypocapnia**	**Hypercapnia** + normoxaemia or hypoxaemia
PaO_2	⭐ **↓ (< 60)**	Normal or ↓ (< 80)
$PaCO_2$	⭐ **Normal or ↓ (< 33)**	⭐ **↑ (> 50)**

⚠️ **Slide inconsistency:** slide 5 defines hypercapnic failure as **$PaCO_2$ > 45**, slide 6 as **> 50**. Most texts use **> 45 (or ≥ 50) mmHg with pH < 7.35** for acute type 2; know both numbers.

[added] **Acute vs chronic type 2:**

	Acute	**Chronic**	**Acute-on-chronic**
pH	↓	Near-normal	↓
HCO_3^-	Normal	↑ (renal compensation)	↑, but pH still low

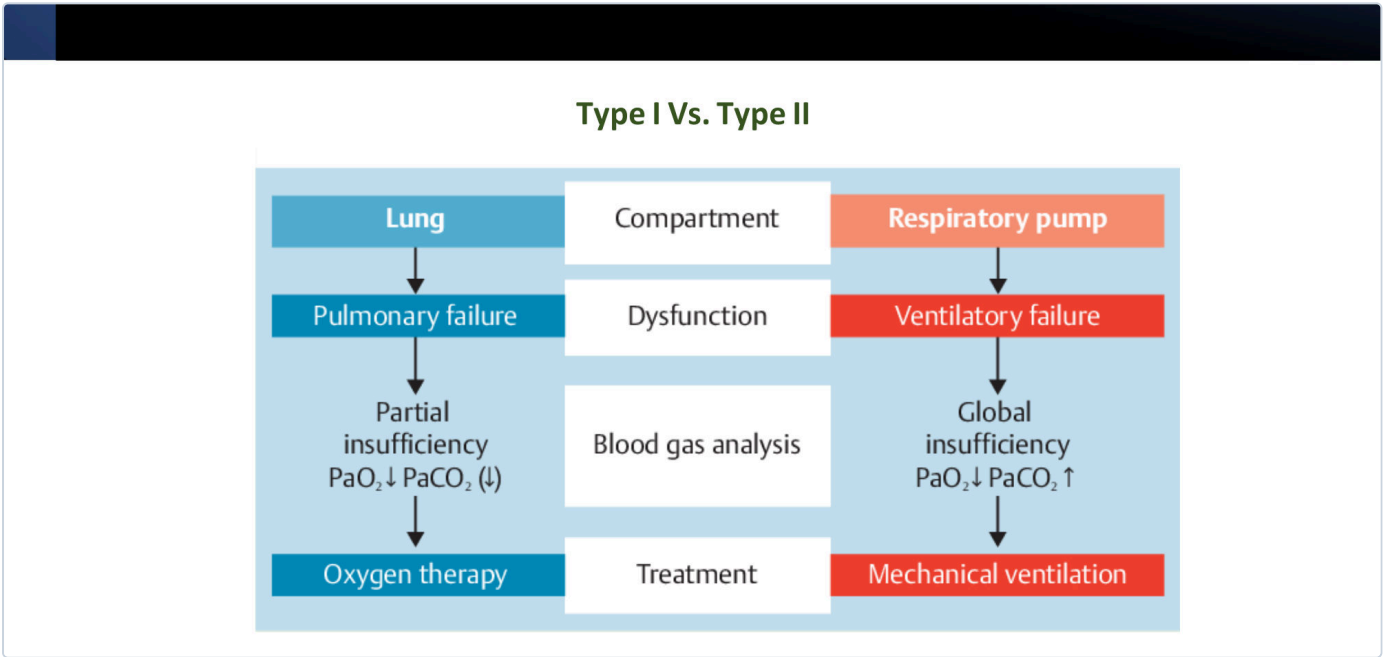


Slide 6: Classification of respiratory failure

2.2 Pathophysiology: Lung vs Pump (Slides 7–8) ⭐

- **Type I:**
  1. pulmonary disease (asthma, pneumonia) → an under-ventilated area is still perfused → **hypoxic, CO_2-rich blood** enters the circulation
  2. ↑ ventilation of neighbouring normal lung → **↑ CO_2 excretion, correcting CO_2** to normal
  3. ⭐ but it **cannot correct O_2**, because Hb passing through normal areas is **already fully saturated** (*[added]* the sigmoid O_2–Hb curve vs the linear CO_2 curve)
- **Type II:** disease (acute severe asthma, **AECOPD**) → **severe V/Q mismatch** → not enough functioning lung to clear CO_2 → **↑ CO_2**.

	Lung (Type 1)	Respiratory pump (Type 2)
Dysfunction	**Pulmonary failure**	**Ventilatory failure**
Blood gas	**Partial insufficiency**: PaO_2 ↓, $PaCO_2$ (↓)	**Global insufficiency**: PaO_2 ↓, $PaCO_2$ ↑
Treatment	⭐ **Oxygen therapy**	⭐ **Mechanical ventilation** (*[added]* NIV first if appropriate)

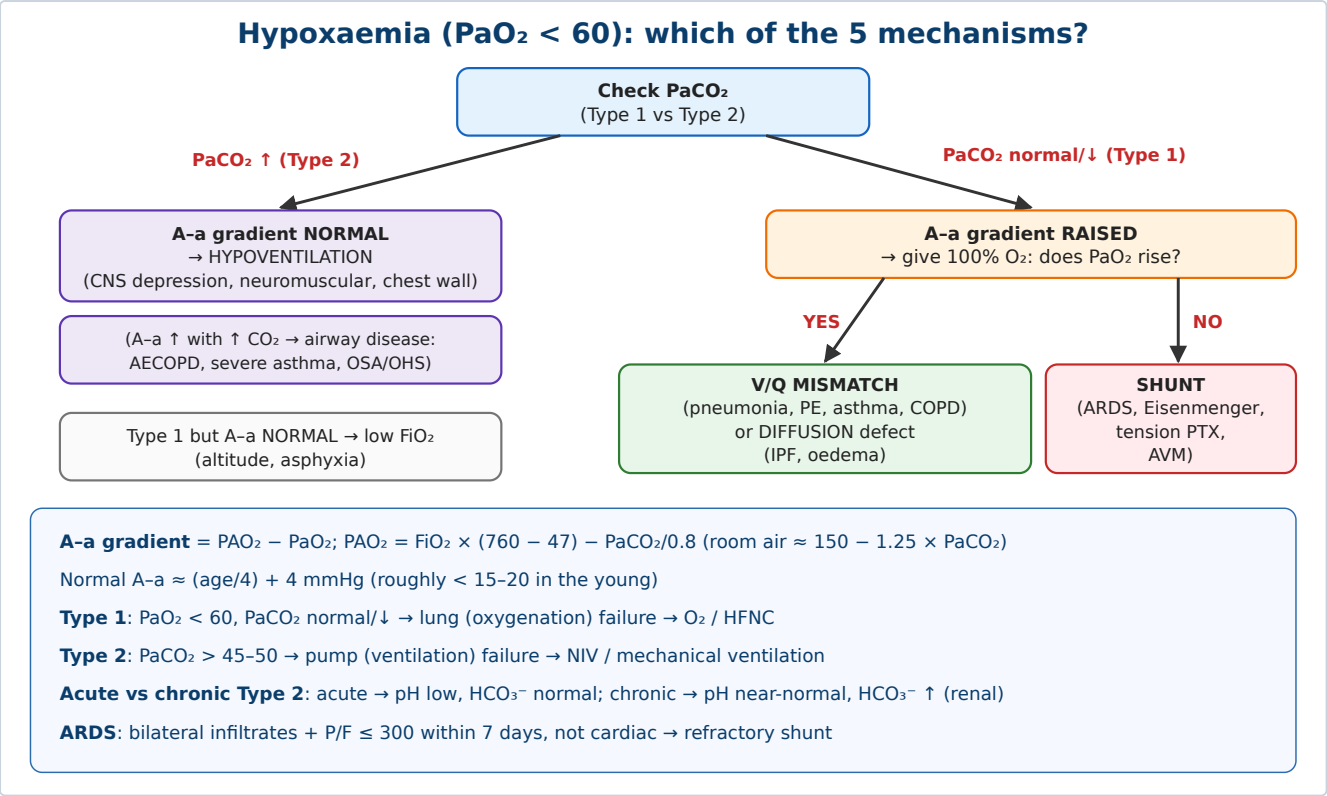


Slide 8: Type I vs Type II (lung vs pump)

2.3 Aetiology (Slides 9–10) ⭐⭐

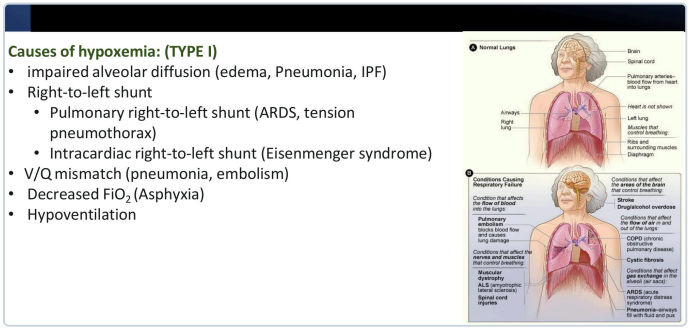
Causes of hypoxaemia (Type I)	Causes of hypercapnia (Type II)
Impaired alveolar diffusion: oedema, pneumonia, **IPF**	**Airway obstruction / ↑ dead space**: **AECOPD, asthma**, drowning, OSA
⭐ **Right-to-left shunt**: **pulmonary** (**ARDS**, tension pneumothorax) or **intracardiac** (**Eisenmenger**)	⭐ **CNS depression**: **opioid** intoxication, **benzodiazepines**
⭐ **V/Q mismatch**: pneumonia, **embolism**	↓ **Chest-wall compliance**: rib fracture (*[added]* flail chest), ankylosing spondylitis, (slide also lists sleep apnoea)
↓ **FiO_2** (asphyxia, *[added]* altitude)	⭐ **Respiratory muscle weakness**: **myasthenia gravis, Guillain–Barré**, myopathies, **ALS**, **high cervical cord injury**, poliomyelitis
Hypoventilation	**Electrolyte disturbances** (*[added]* ↓ K, ↓ PO_4, ↓ Mg)

[added] **Hypoventilation** causes hypoxaemia, but it is the hallmark of **type 2** (with a **normal A-a gradient**). See the added diagram.

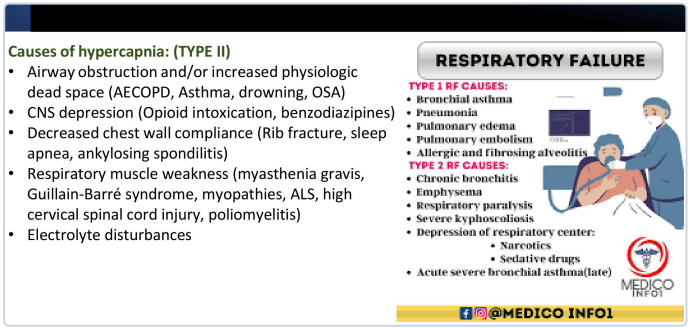


Added diagram: Five mechanisms of hypoxaemia and the A–a gradient

Mechanism *[added]*	A-a gradient	Response to 100% O_2	Example
↓ FiO_2	Normal	Corrects	Altitude
Hypoventilation	**Normal**	Corrects	Opioid overdose, neuromuscular disease
Diffusion defect	↑	Corrects	IPF
V/Q mismatch (commonest)	↑	**Corrects**	Pneumonia, PE, COPD
⭐ **Shunt**	↑	⭐ **Does NOT correct**	**ARDS**, Eisenmenger, AVM



Slide 9: Causes of hypoxaemia



Slide 10: Causes of hypercapnia

2.4 Clinical Features (Slides 11–13) ⭐

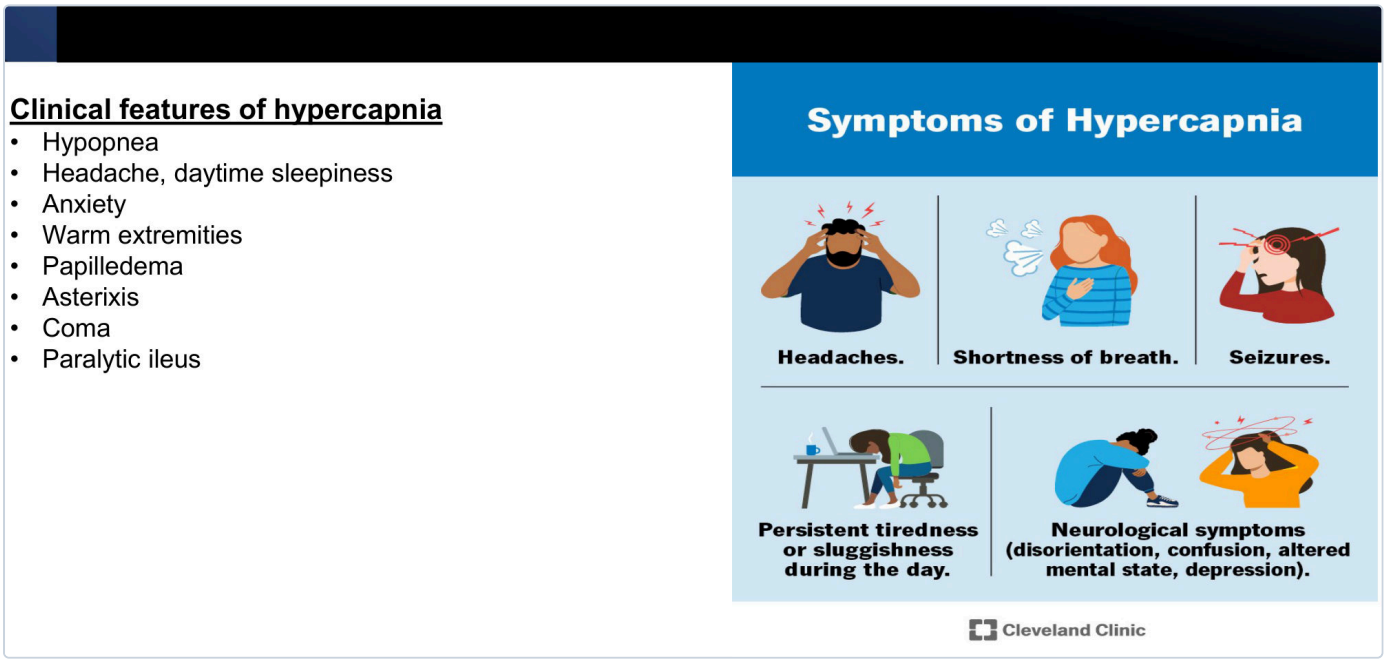
Hypoxaemia	Hypercapnia
Tachypnoea, dyspnoea	**Hypopnoea**
⭐ **Cyanosis**	⭐ **Headache**, daytime sleepiness

Hypoxaemia	Hypercapnia
Pleuritic chest pain	Anxiety
Tachycardia, arrhythmia	⭐ **Warm extremities** (vasodilatation; *[added]* bounding pulse)
Altered mental status: confusion, somnolence, agitation	⭐ **Papilloedema**
—	⭐ **Asterixis (flapping tremor)**
—	⭐ **Coma** (*[added]* CO_2 narcosis)
—	**Paralytic ileus**

Signs of respiratory distress (slide 13):

- ⭐ **inability to speak in full sentences**
- ↑ work of breathing: **nasal flaring, grunting, pursed lips**
- **accessory muscle use**
- tachypnoea/hyperventilation, tachycardia, cyanosis
- ⭐ **impaired mental state** (fatigue, hypercapnia, or a CNS cause)

[added] A **falling respiratory rate or drowsiness in a tiring patient = impending arrest**, not improvement.



Slide 12: Symptoms of hypercapnia

2.5 Diagnosis and Management (Slides 14–16) ⭐

Diagnosis: clinical; ABG, rapid labs and bedside imaging identify the type and guide treatment.

- ⭐ **ABG**
- **CXR or CT chest**
- **CBC and metabolic panel**
- **Cardiac enzymes** if cardiogenic oedema is suspected
- **Cause:** CXR, **POCUS**, CT, **CTA**

Approach (slide 15):

1. ⭐ **ABCDE**

2. **immediate respiratory support** according to cause and severity
3. ⭐ **identify and treat rapidly reversible causes**: **tension pneumothorax, pulmonary oedema, anaphylaxis, status asthmaticus, foreign-body aspiration**
4. focused evaluation; initial tests (ABG, labs, **ECG, CXR, POCUS**)
5. advanced tests as needed
6. **treat the cause**

Mnemonic *[added]* for the rapidly reversible causes: "PEAS-F":

- **P**neumothorax (tension)
- **E**dema (pulmonary)
- **A**naphylaxis
- **S**tatus asthmaticus
- **F**oreign body

Monitoring and disposition	Respiratory support
ICU admission for respiratory/haemodynamic support	**Secure the airway**
Monitor respiratory and haemodynamic parameters	**Rescue breaths** immediately in **respiratory arrest**
⭐ **Serial ABGs**	**O_2 therapy**: slide threshold **SpO_2 < 95%** (see note); consider ⭐ **high-flow nasal cannula** for hypoxaemic failure
Repeat labs/imaging if the status changes	**Mechanical ventilation**: consider indications for **NIPPV** vs **invasive**

⚠️ *[added]* O_2 targets: **94–98%** for most patients, **88–92%** in those at risk of hypercapnia (COPD, OHS, neuromuscular). Excess O_2 in type 2 failure worsens CO_2 retention.

[added] **NIV (BiPAP)** is best evidenced for **AECOPD with pH 7.25–7.35** and **cardiogenic pulmonary oedema**.

- **Intubate if:**
  - NIV fails or is contraindicated
  - GCS ≤ 8 / unable to protect the airway
  - haemodynamic instability
  - respiratory arrest
  - severe acidosis



Slide 16: Monitoring and respiratory support

🔁 **Recap: Respiratory failure.**

- **Type 1:** PaO_2 < 60, CO_2 normal/↓. Lung failure: V/Q, shunt, diffusion, ↓ FiO_2 → **O_2/HFNC**.
- **Type 2:** $PaCO_2$ > 45–50. Pump failure: COPD/asthma, opioids/benzodiazepines, chest wall, neuromuscular → **ventilation (NIV/IMV)**.
- **Hypercapnia signs:** headache, warm peripheries, papilloedema, asterixis, coma.
- **ABCDE**; treat reversible causes (tension PTX, oedema, anaphylaxis, asthma, foreign body); serial ABGs; ICU.

▶️ **Up next: ARDS.**

2.6 ARDS: Definition and Causes (Slides 17–19) ⭐⭐

Definition:

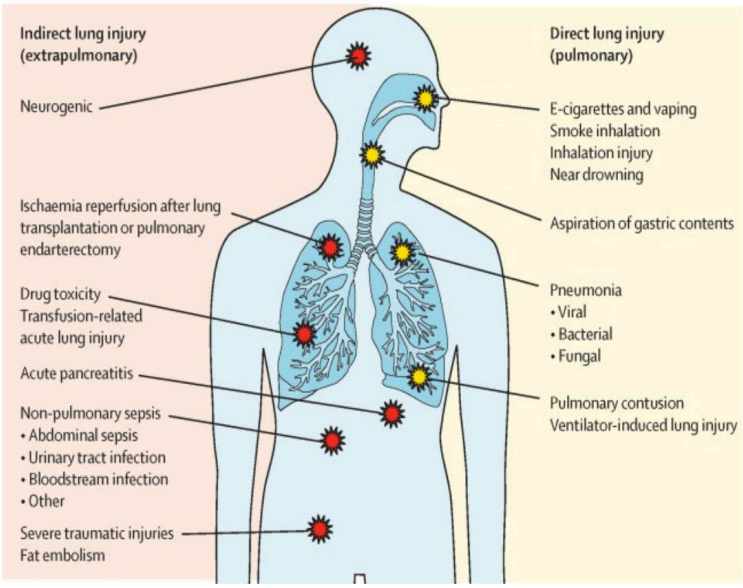
- **acute respiratory failure** with **hypoxaemia** and **bilateral pulmonary infiltrates**
- ⭐ **not fully explained by heart failure or fluid overload**
- **not a primary disease**: it arises from other conditions → a **widespread inflammatory process**

Systemic (indirect) causes	Direct lung injury
⭐ **Sepsis (most common cause)**	⭐ **Pneumonia**
Trauma	⭐ **Aspiration**
Shock	Inhaled toxins; inhalation injury (e.g. hyperbaric O_2)
⭐ **Massive transfusion (TRALI)**	**Pulmonary contusion**
⭐ **Acute pancreatitis**	**Drowning**
Haematopoietic stem-cell transplantation	⭐ **Fat embolism** (*[added]* long-bone fractures)
Medications	**Amniotic fluid embolism**
Recreational drugs (cocaine)	Lung transplantation

(ARDS)

Systemic Causes:

- Sepsis (most common cause)
- Trauma
- Shock
- Massive blood transfusion (TRALI)
- Acute pancreatitis
- Hematopoietic stem cell transplantation
- Medications
- Recreational drug overdose (e.g., cocaine)



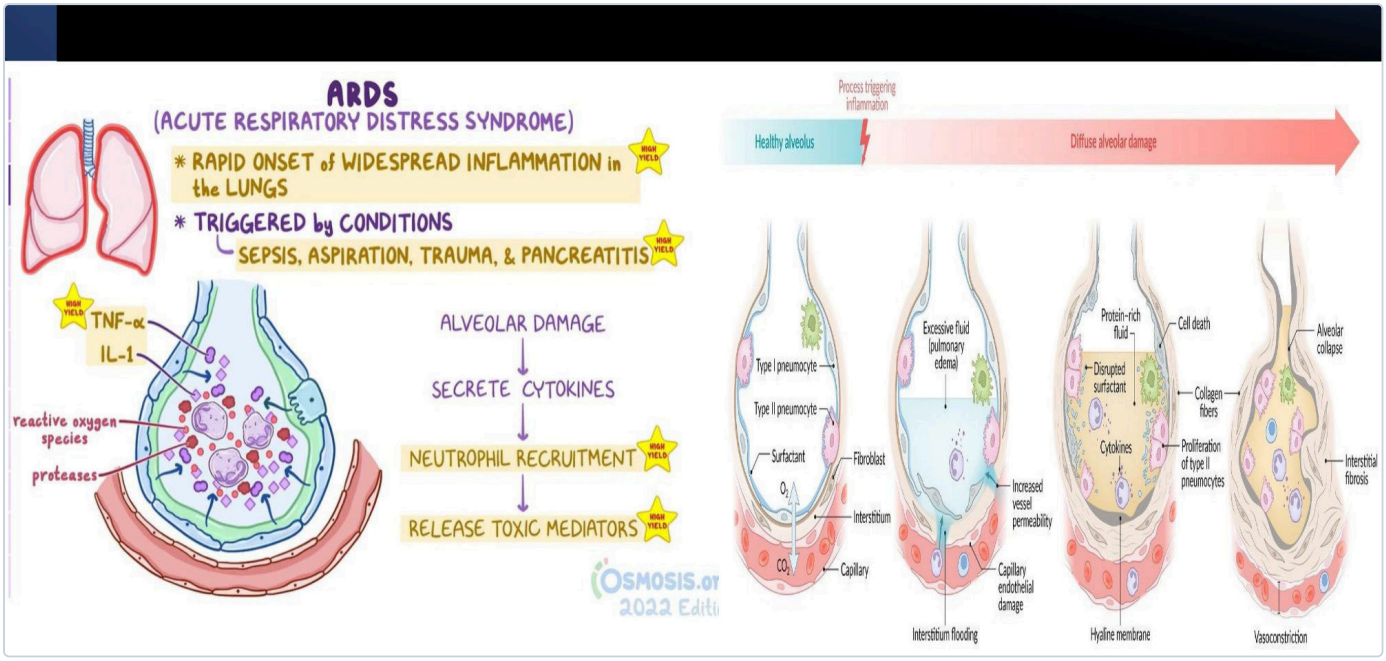
Slide 18: Systemic causes of ARDS

2.7 ARDS: Pathophysiology (Slides 20–21) ⭐⭐

Trigger: tissue damage (pulmonary or extrapulmonary) → **inflammatory mediators (TNF-α, IL-1)** → ⭐ **neutrophil sequestration in the alveoli** → reactive oxygen species and proteases → injury to alveolar capillaries and endothelium = ⭐ **diffuse alveolar damage (DAD)**.

Phase	Events
⭐ **Exudative**	Fluid floods the interstitium and alveoli → ⭐ **non-cardiogenic pulmonary oedema with NORMAL PCWP** → ↓ **compliance**, respiratory distress
⭐ **Hyaline membranes**	Neutrophils + **protein-rich** fluid → **hyaline membranes** → impaired gas exchange → hypoxaemia → hyperventilation → ⭐ **respiratory alkalosis** (early)
Vascular	Hypoxaemia → hypoxic pulmonary vasoconstriction → **pulmonary hypertension**
Surfactant loss	Damage to **type I and II pneumocytes** → ↓ surfactant → **alveolar collapse** → ⭐ **intrapulmonary shunting** (refractory hypoxaemia)

Phase	Events
Organising (late)	**Type II pneumocyte proliferation** + **fibroblasts** → ⭐ **progressive interstitial fibrosis**



Slide 20: ARDS pathophysiology (DAD)

(ARDS)

Pathophysiology:

•Tissue damage (pulmonary or extrapulmonary) → release of inflammatory mediators → **sequestration of** neutrophils into alveoli → injury to alveolar capillaries and endothelial cells (**diffuse alveolar damage, DAD**) leading to:

- **Exudative phase**: excess fluid in interstitium and on alveolar surface → pulmonary edema with normal pulmonary capillary wedge pressure (**noncardiogenic pulmonary edema**) → **decreased lung compliance and respiratory distress**
- **Hyaline membrane formation**: exudation of neutrophils and protein-rich fluid into the alveolar space → formation of alveolar **hyaline membranes** → impaired gas exchange → hypoxemia →hyperventilation → respiratory alkalosis
  - Hypoxemia → chronic hypoxic pulmonary vasoconstriction → pulmonary hypertension
  - **Damage** to type I and type II pneumocytes → decrease in surfactant → alveolar collapse → intrapulmonary shunting
- **Organizing phase (late stage)**: **proliferation of type II pneumocytes** and infiltration of fibroblasts → **progressive interstitial fibrosis**

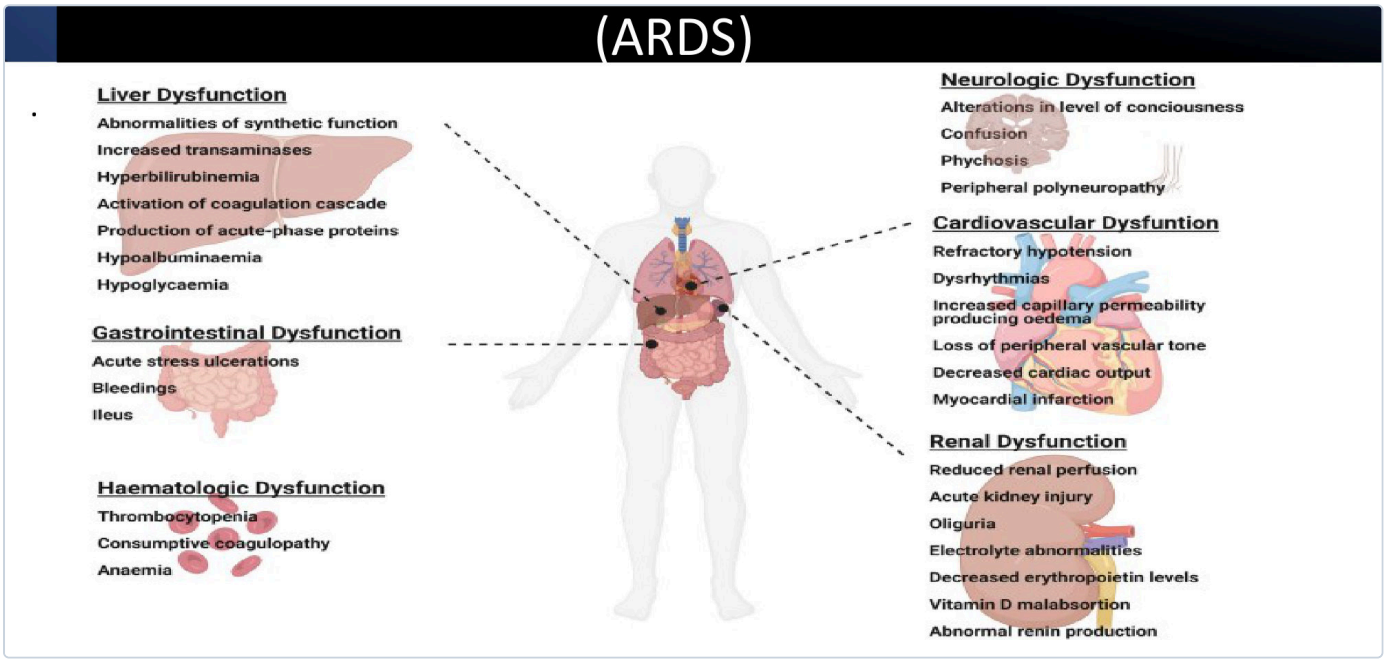
Slide 21: ARDS phases

2.8 ARDS: Clinical Features and Multi-Organ Effects (Slides 22–23)

Symptoms and signs:

- **acute dyspnoea**; tachypnoea, tachycardia; **cyanosis**
- ⭐ **diffuse crackles**
- ± fever, cough, chest pain
- *[added]* onset typically **12–72 h** after the insult

Organ dysfunction (slide 23)	Manifestations
Liver	↓ synthetic function, ↑ transaminases, ↑ bilirubin, coagulation activation, acute-phase proteins, ↓ albumin, hypoglycaemia
GI	**Stress ulcers**, bleeding, **ileus**
Haematological	Thrombocytopenia, **consumptive coagulopathy (DIC)**, anaemia
Neurological	↓ consciousness, confusion, psychosis, polyneuropathy
Cardiovascular	Refractory hypotension, arrhythmia, capillary leak, ↓ vascular tone, ↓ cardiac output, MI
Renal	↓ perfusion, **AKI**, oliguria, electrolyte disturbance, ↓ EPO, ↓ vitamin D activation, abnormal renin



Slide 23: Multi-organ dysfunction in ARDS

2.9 ARDS: Diagnostic Criteria (Slides 24–25) ⭐⭐

New Global Definition (2023) (the slide is headed "Berlin criteria"; *[added]* Berlin 2012 is the predecessor):

Criterion	Requirement
⭐ **Timing**	**Within 7 days** of a known risk factor
⭐ **Oedema origin**	**Not explained by cardiogenic oedema or volume overload**
⭐ **Imaging**	**Bilateral infiltrates on CXR or CT**, *or* on **lung ultrasound** (by a trained operator); not explained by nodules, effusion or atelectasis
⭐ **Hypoxaemia (PaO_2/FiO_2)**	**Mild 201–300** (on NIV/CPAP PEEP ≥ 5, **or HFNO ≥ 30 L/min**) · **Moderate 101–200** (PEEP ≥ 5) · **Severe ≤ 100** (PEEP ≥ 5)
SpO_2/FiO_2 (new)	**≤ 315 with SpO_2 ≤ 97%** (no ABG needed)

[added] **What 2023 changed from Berlin:**

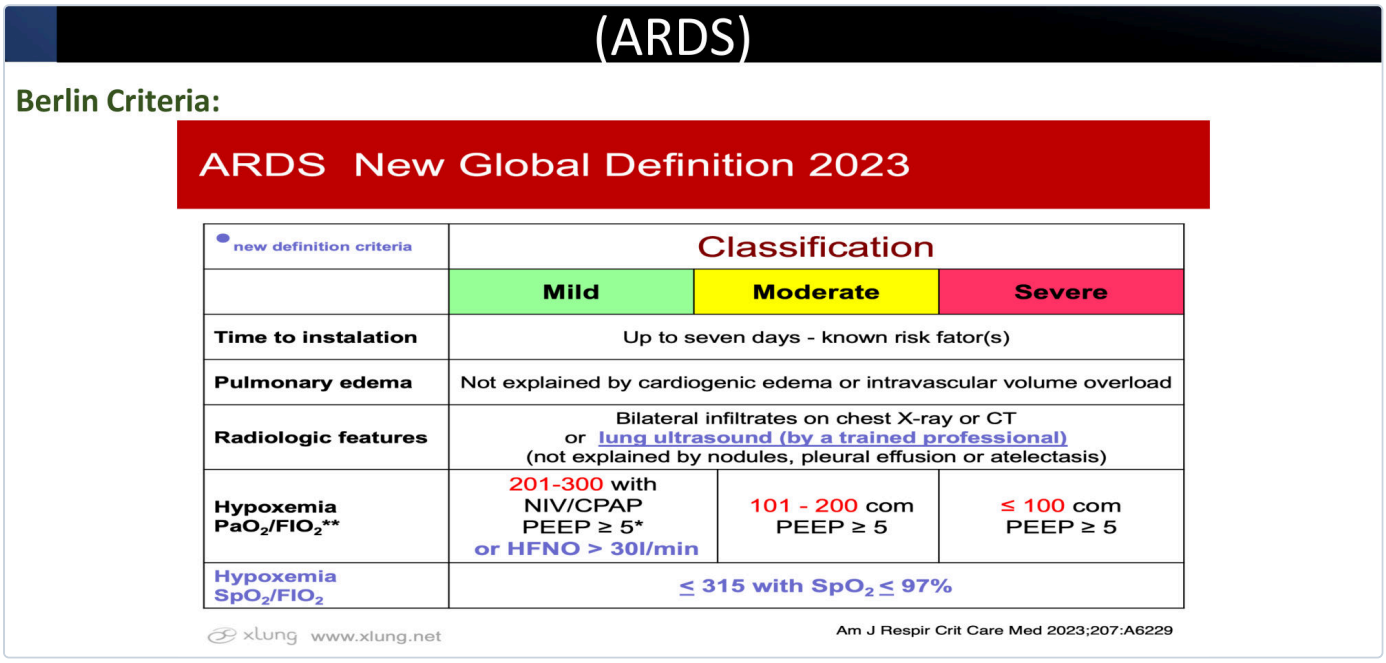
- **HFNO** patients are included (non-intubated ARDS)
- **SpO_2/FiO_2** allowed
- **ultrasound** allowed for imaging
- a modification for **resource-limited settings** (no PEEP or O_2-flow requirement)

Diagnostics (slide 25):

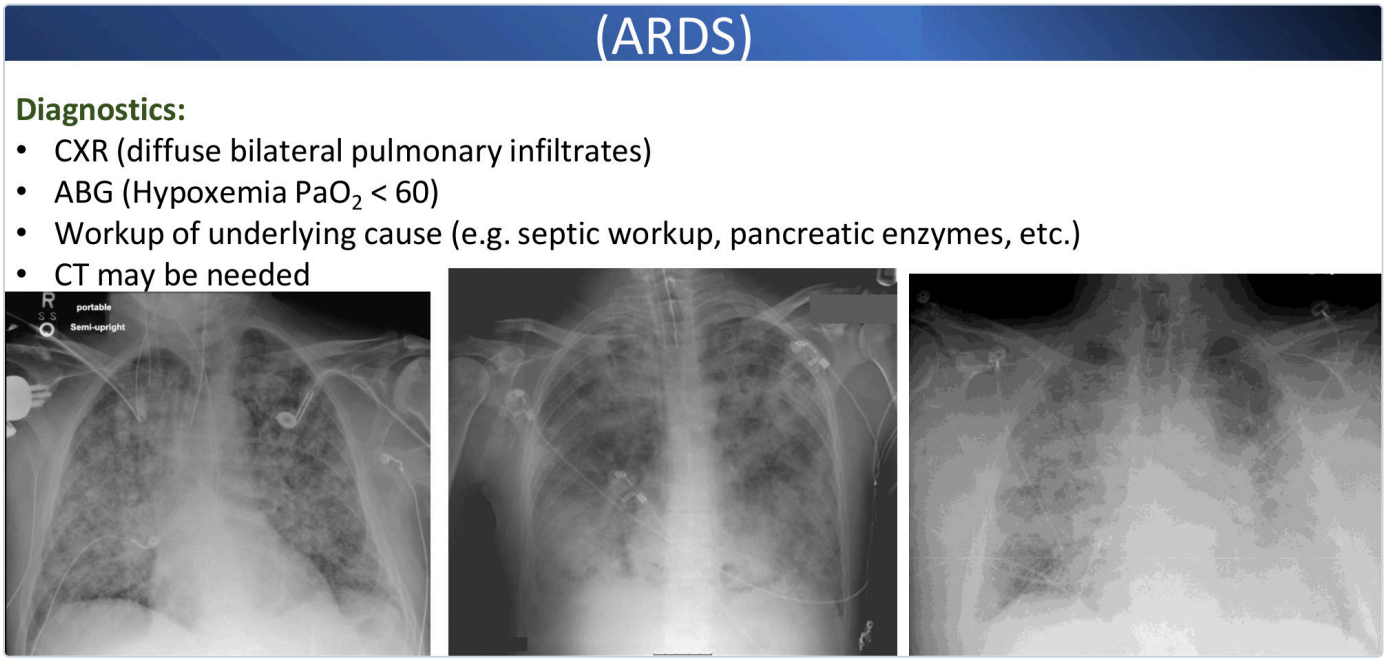
- **CXR:** ⭐ diffuse **bilateral** infiltrates (*[added]* "white-out", often sparing the costophrenic angles; no cardiomegaly)
- **ABG:** hypoxaemia (slide: "PaO_2 < 60")
- **Work-up of the cause:** septic screen, **pancreatic enzymes**
- **CT** if needed

⚠️ ARDS severity is defined by the **P/F ratio** (with PEEP), not by "PaO_2 < 60".

[added] **BNP / echo** help exclude a cardiogenic cause: normal BNP favours ARDS (lecture case 1).



Slide 24: ARDS Global Definition 2023



Slide 25: CXR of ARDS

2.10 ARDS: Management (Slide 26) ⭐⭐

General:

- ⭐ **early ICU admission and treat the underlying cause**
- **oxygenation**
- **glucocorticoids in intubated patients** (*[added]* ATS 2024: conditional recommendation)
- ⭐ **conservative fluids (diuretics)** once shock resolves

⭐ Lung-protective ventilation (ARDSNet):

Setting	Target ***[added numbers]***
Low tidal volume	**6 mL/kg predicted body weight** (range 4–8)
Low plateau pressure	**≤ 30 cmH_2O**
Higher PEEP	To recruit alveoli and reduce shunt (PEEP/FiO_2 table)
[added] **Permissive hypercapnia**	pH ≥ 7.20–7.25 acceptable
[added] **Oxygen target**	PaO_2 55–80 mmHg / SpO_2 88–95%

By severity:

Severity	Add
Moderate or severe	⭐ **Prone positioning** (*[added]* if P/F < 150, ≥ 12–16 h/day; PROSEVA) and **high-PEEP strategy**
Early severe	Consider **neuromuscular blockers**
Severe, refractory	⭐ **ECMO** (VV-ECMO; *[added]* the slide image shows VA-ECMO)

[added] **Predicted body weight:**

- **men** = 50 + 0.91 × (height cm − 152.4)
- **women** = 45.5 + 0.91 × (height cm − 152.4)

Mnemonic *[added]* for ARDS ventilation, "6-30-PEEP-Prone": **6** mL/kg PBW · plateau ≤ **30** · higher **PEEP** · **Prone** if P/F < 150.

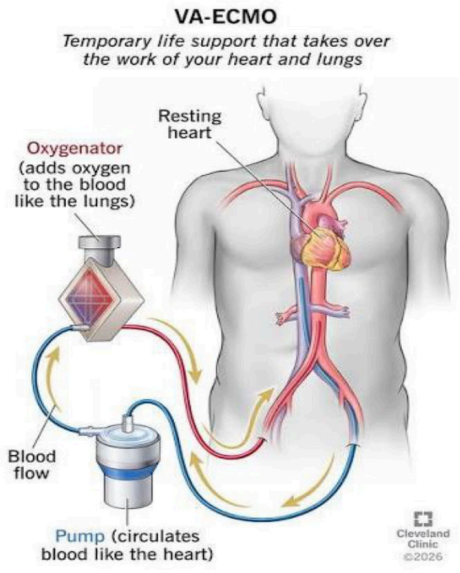
(ARDS)

Management:

- Early ICU admission & Treat the underlying cause
- Oxygenation
- Glucocorticoids in intubated patients
- Lung-protective ventilation per the ARDSNet study:
  - Low tidal volume
  - Low plateau pressure
  - High positive end expiratory pressure (PEEP)

Adjust therapy based on severity.

  - Moderate or severe: Use prone positioning and high PEEP strategy.
  - Early severe ARDS: Consider neuromuscular blockers.
  - Severe ARDS not responding to therapeutic adjustments: Consider ECMO.
- Fluid management (diuretics)



Slide 26: ARDS management

> 🔁 **Recap: ARDS.**
> - Hypoxaemia + bilateral infiltrates within 7 days, not cardiac.
> - Causes: sepsis (commonest), pneumonia, aspiration, pancreatitis, trauma, TRALI, fat embolism.
> - Neutrophil-mediated DAD → protein-rich oedema (normal PCWP) → hyaline membranes → surfactant loss → **shunt** → fibrosis.
> - 2023 definition: P/F 201–300 / 101–200 / ≤ 100; S/F ≤ 315; HFNO allowed.
> - **6 mL/kg PBW, plateau ≤ 30, high PEEP, prone, NMB, ECMO, conservative fluids, steroids.**
>
> ▶️ **Up next: Cyanosis.**

2.11 Approach to Cyanosis (Slides 27–28) ⭐

History:

- **onset** (age; acute vs chronic)
- **chest pain**, **dyspnoea**
- **smoking**
- lung disease or HF
- **allergies**
- **medications (β-blockers**: *[added]* peripheral cyanosis; and nitrates/dapsone/local anaesthetics → methaemoglobinaemia)

Examination: vital signs, **pulse oximetry**, heart, lungs, extremities (**clubbing, cyanosis, oedema**).

Work-up:

- **General: SpO_2, ABG, CBC, ECG, CXR**; **echo** if a cardiac cause is suspected.
- **Specific:** sputum and blood cultures; **V/Q scan and CTA**; **TTE**; **duplex Doppler**.

Central cyanosis (tongue, lips)	Peripheral cyanosis (fingers, toes)
↓ **O_2 saturation:** severe respiratory disease, **pulmonary oedema, PE**, **congenital cyanotic heart disease**	⭐ **All causes of central cyanosis**
⭐ **Abnormal haemoglobin: methaemoglobinaemia, sulphaemoglobinaemia**	**Cold exposure**
—	**Raynaud's phenomenon**
—	**Arterial occlusion**; **venous occlusion**
—	**Acrocyanosis**

[added]

- Cyanosis appears when **deoxygenated Hb > 5 g/dL** (capillary). It is **masked in anaemia** and **exaggerated in polycythaemia**.
- **Methaemoglobinaemia:**
  - **"saturation gap"**: SpO_2 ~85% fixed, but **normal PaO_2**
  - **chocolate-brown blood**
  - treat with **methylene blue** (not in **G6PD** deficiency)
- **Peripheral cyanosis with warm hands → think central; cold hands → peripheral.**

General Workup:

- Oxygen saturation
- ABG
- CBC
- ECG
- CXR
- Echo if Cardiac cause is suspected

Specific Workup:

- Sputum & blood culturs
- V/Q scan & CTA
- TTE
- Duplex Doppler

DDx:

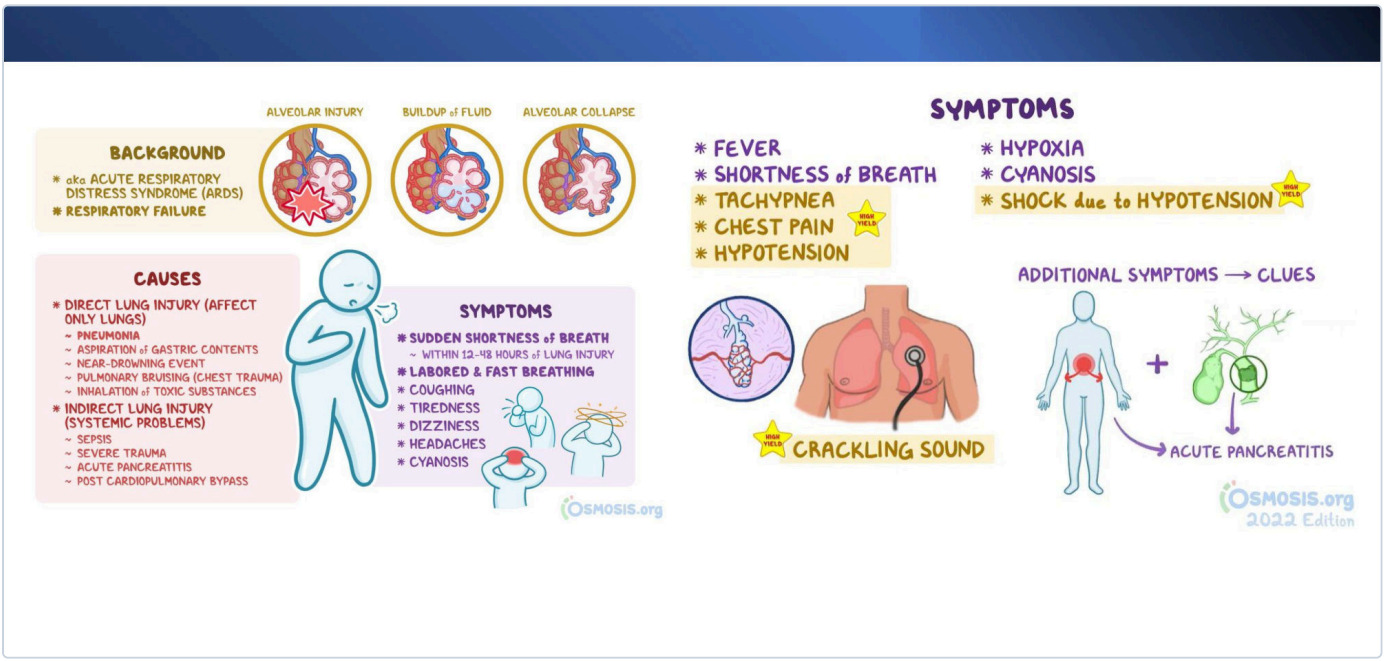
CENTRAL CYANOSIS

DECREASED OXYGEN SATURATION
- Severe respiratory disease
- Pulmonary oedema
- Pulmonary embolism
- Congenital cyanotic heart disease

ABNORMAL HAEMOGLOBIN
- Methaemoglobinaemia
- Sulphaemoglobinaemia

PERIPHERAL CYANOSIS
- All causes of central cyanosis
- Cold exposure
- Raynaud's phenomenon
- Arterial occlusion
- Venous occlusion
- Acrocyanosis

Slide 28: Cyanosis work-up and differential



Slide 29: ARDS summary (Osmosis)

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3. High-Yield Comparison Tables

Table A: Type 1 vs Type 2 Respiratory Failure

	Type 1	Type 2
Problem	**Oxygenation** (lung)	**Ventilation** (pump)
PaO_2	**< 60**	Normal or ↓
$PaCO_2$	Normal/↓ (< 33)	**> 45–50**
A–a gradient	↑ (except ↓ FiO_2)	Normal (pure hypoventilation) or ↑ (airway disease)

	Type 1	Type 2
Causes	Pneumonia, **ARDS**, PE, oedema, IPF, shunt	**AECOPD**, asthma (late), **opioids/benzodiazepines**, **MG/GBS/ALS**, chest wall, OHS
Clues	Cyanosis, tachypnoea, confusion	**Headache, warm peripheries, papilloedema, asterixis**, drowsiness, coma
Treatment	**O_2 / HFNC** (CPAP)	**NIV / invasive ventilation**; controlled O_2 88–92%

Table B: ARDS vs Cardiogenic Pulmonary Oedema *[added synthesis; slides 17, 21, 30]*

	ARDS	Cardiogenic oedema
Mechanism	↑ **Permeability** (alveolocapillary leak)	↑ **Hydrostatic** (capillary pressure)
PCWP	**Normal (≤ 18)**	**↑ (> 18)**
BNP	Normal/mildly ↑	**High**
Oedema fluid	**Protein-rich** exudate	Transudate
CXR	Bilateral, peripheral, no cardiomegaly, no effusions	**Cardiomegaly**, upper-lobe diversion, Kerley B lines, effusions
O_2 response	**Refractory** (shunt)	Responds; diuretics work

Table C: Global Definition 2023 ARDS Severity

Severity	P/F (PEEP ≥ 5 or HFNO ≥ 30 L/min)	Extra therapy
Mild	201–300	Lung-protective ventilation
Moderate	101–200	+ **high PEEP**, **prone** (if P/F < 150)
Severe	≤ 100	+ **NMB** (early), **prone**, **ECMO** if refractory

Table D: Direct vs Indirect ARDS

Direct (pulmonary)	Indirect (systemic)
Pneumonia, aspiration, contusion, drowning, inhalation, **fat embolism**, **amniotic fluid embolism**, lung transplant	**Sepsis**, trauma, shock, **TRALI**, **pancreatitis**, HSCT, drugs, **cocaine**

Table E: Central vs Peripheral Cyanosis

	Central	Peripheral
Site	**Tongue, lips**, mucosa + extremities	Fingers, toes, nose only
SpO_2	↓ (except MetHb, which shows a gap)	Normal
Extremities	Warm	**Cold**
Causes	Lung disease, oedema, PE, cyanotic CHD, **MetHb/SulfHb**	Cold, **Raynaud's**, arterial/venous occlusion, acrocyanosis, ↓ cardiac output, β-blockers

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4. Rules, Exceptions, Patterns

- **Hypoxaemic failure: PaO_2 < 60; hypercapnic: $PaCO_2$ > 45** (slide 6 says > 50).
- **Type 1 compensates CO_2 but not O_2** (Hb already saturated in normal units).

- **Shunt doesn't correct with 100% O_2**; V/Q mismatch does.
- **Hypoventilation = normal A–a gradient** (opioids, neuromuscular).
- **Hypercapnia: warm peripheries, papilloedema, asterixis, coma.**
- **Reversible causes first:** tension PTX, oedema, anaphylaxis, status asthmaticus, foreign body.
- **Controlled O_2 (88–92%)** if at risk of CO_2 retention.
- **ARDS = not cardiogenic** (normal PCWP/BNP); **sepsis is the commonest cause**.
- **Early ARDS → respiratory alkalosis**; late → fibrosis.
- **ARDS severity = P/F with PEEP ≥ 5** (2023: HFNO ≥ 30 L/min also counts; S/F ≤ 315).
- **Lung-protective ventilation: 6 mL/kg PBW, plateau ≤ 30, PEEP; prone if P/F < 150; NMB early severe; ECMO refractory.**
- **Conservative fluids** in ARDS once shock resolves.
- **Cyanosis needs > 5 g/dL deoxyHb** → hidden by anaemia; **MetHb = saturation gap**.

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5. Clinical Correlations

- **Opioid/benzodiazepine overdose** → type 2 failure → **naloxone** (opioids), airway support (Substance Use lecture).
- **Myasthenic crisis / GBS** → monitor **FVC** (< 15–20 mL/kg → intubate) *[added]*; ABG changes are late.
- **AECOPD** → NIV for pH 7.25–7.35; controlled O_2 88–92%.
- **OHS** → acute-on-chronic type 2 failure; ↑ HCO_3^- (Sleep Apnoea lecture).
- **Acute pancreatitis** (alcohol) → ARDS (lecture case 1); periumbilical haematoma = **Cullen's sign**.
- **Massive transfusion** → **TRALI** within 6 h *[added]*.
- **Long-bone fracture** → **fat embolism**: hypoxaemia + petechiae + confusion *[added]*.
- **Sepsis** (upcoming lecture) → the commonest cause of ARDS.
- **Dapsone/nitrates/local anaesthetics** → **methaemoglobinaemia**: cyanosis with a normal PaO_2 *[added]*.
- **Eisenmenger** → intracardiac shunt; cyanosis not corrected by O_2.

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6. Rapid Revision (Last-Day)

1. **Respiratory failure:** can't maintain O_2/CO_2. **Hypoxaemic PaO_2 < 60; hypercapnic $PaCO_2$ > 45** (slide 6: > 50). Respiratory arrest = no breathing with a pulse.
2. **Acute** (minutes–hours) vs **chronic** (COPD, ILD, OHS).
3. **Type 1:** PaO_2 < 60, $PaCO_2$ normal/↓ (< 33). **Type 2:** $PaCO_2$ ↑, PaO_2 normal/↓.
4. **Type 1 mechanism:** normal units blow off CO_2 but can't add O_2 (Hb saturated). **Type 2:** severe V/Q mismatch / pump failure.
5. **Lung failure → O_2; pump failure → mechanical ventilation.**
6. **Type 1 causes:**
   - diffusion (oedema, pneumonia, IPF)
   - **shunt** (ARDS, tension PTX, Eisenmenger)
   - **V/Q** (pneumonia, PE)
   - ↓ FiO_2
   - hypoventilation
7. **Type 2 causes:**
   - airway/dead space (AECOPD, asthma, drowning, OSA)
   - **CNS depression (opioids, benzodiazepines)**
   - chest wall (rib fracture, AS)

    - **muscle weakness (MG, GBS, ALS, high cord injury, polio)**
    - electrolytes
8. **Hypoxaemia features:** tachypnoea, dyspnoea, cyanosis, tachycardia/arrhythmia, confusion.
9. **Hypercapnia features:** hypopnoea, **headache**, sleepiness, anxiety, **warm extremities, papilloedema, asterixis, coma**, ileus.
10. **Distress:** can't finish sentences, nasal flaring, grunting, pursed lips, accessory muscles, ↓ mental state.
11. **Work-up:** ABG, CXR/CT, CBC/metabolic panel, troponin if cardiac; POCUS, CTA.
12. **Approach: ABCDE** → support → **reversible causes (tension PTX, oedema, anaphylaxis, status asthmaticus, foreign body)** → tests → treat the cause → ICU, serial ABGs.
13. **O_2:** slide threshold SpO_2 < 95%; **HFNC** for type 1; NIPPV vs invasive. *[added]* Targets 94–98% (88–92% if CO_2 retainer).
14. **ARDS:** hypoxaemia + bilateral infiltrates, not HF/fluid; a syndrome secondary to another condition.
15. **ARDS causes:** - **systemic: sepsis** (commonest), trauma, shock, **TRALI**, **pancreatitis**, HSCT, drugs, cocaine - **direct: pneumonia, aspiration**, toxins, contusion, drowning, **fat/amniotic embolism**, lung transplant
16. **ARDS pathology:** mediators → neutrophils → **DAD** → exudative (non-cardiogenic oedema, **normal PCWP**, ↓ compliance) → **hyaline membranes** → hypoxaemia → **respiratory alkalosis**; ↓ surfactant → collapse → **shunt**; PH; **organising → fibrosis**.
17. **ARDS signs:** acute dyspnoea, tachypnoea, tachycardia, cyanosis, **diffuse crackles**; multi-organ failure.
18. **Global Definition 2023:** within **7 days**; not cardiac; bilateral infiltrates on CXR, CT or **US**; P/F **201–300 / 101–200 / ≤ 100** (PEEP ≥ 5 or **HFNO ≥ 30**); **S/F ≤ 315** (SpO_2 ≤ 97%).
19. **ARDS Rx:** - ICU; treat the cause; O_2 - **steroids** (intubated) - **low TV (6 mL/kg PBW), plateau ≤ 30, high PEEP** - moderate/severe → **prone + high PEEP**; early severe → **NMB**; refractory → **ECMO** - **diuretics** (conservative fluids)
20. **Cyanosis:** - **central** (↓ SaO_2: lung disease, oedema, PE, cyanotic CHD; **MetHb, SulfHb**) - **peripheral** (all central causes + cold, **Raynaud's**, arterial/venous occlusion, acrocyanosis) - work-up: SpO_2, ABG, CBC, ECG, CXR, echo; then cultures, V/Q/CTA, TTE, Doppler

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7. Exam-Style Questions

Q1 (lecture case 1, slide 30). A 63-year-old woman with 20 years of alcohol use disorder has **2 days of severe epigastric pain and nausea**. **9 hours after admission** she becomes dyspnoeic and tachypnoeic; **SpO_2 81% on O_2**. Diffuse crackles, epigastric tenderness, a **periumbilical haematoma**; **normal BNP**; CXR **bilateral lower-zone opacities**. Pathomechanism? **a. Alveolocapillary membrane leakage** · b. Intrapulmonary left-to-right shunt · c. ↑ Surfactant · d. Pulmonary artery embolic obstruction · e. Bronchial mucosal inflammation · f. ↑ Pulmonary capillary pressure

> **Answer: a** *[reasoning added; key not on slides]*.
>
> - **Acute pancreatitis** (alcohol, epigastric pain, **Cullen's sign**) → systemic inflammation → **ARDS**.
> - Neutrophil-mediated **DAD** → **↑ alveolocapillary permeability** → protein-rich non-cardiogenic oedema.
> - Why not the others:
> - **f: normal BNP** argues against ↑ capillary (hydrostatic) pressure, i.e. cardiogenic oedema.
> - **b:** the ARDS shunt is **right-to-left**.
> - **c:** surfactant is **reduced**.

Q2 (lecture case 2, slide 31). A 70-year-old (**180 cm**, 90 kg) with **wet gangrene** develops hypoxaemia on day 3 and is intubated. HR 112, **BP 95/60**, **lactate 4.0**, **creatinine 2.5**, Hb 11.5. Ventilator: **TV 540 mL**, RR 20, **PEEP 5**. ABG: PCO_2 50, **PO_2 55 on FiO_2 1.0**. CXR: new bilateral infiltrates. Next step to improve tissue O_2 delivery? a. Inhaled NO · b. Prone positioning · c. ↑ Tidal volume · **d. ↑ PEEP** · e. RBC transfusion

> **Answer: d** *[reasoning added; key not on slides].*
> - **Septic shock → severe ARDS**: P/F = 55/1.0 = **55 (≤ 100)**.
> - Refractory hypoxaemia from **shunt** improves with **higher PEEP** (recruits collapsed alveoli); slide 26: "**high PEEP**" for moderate/severe ARDS.
> - Why not the others:
> - **c:** wrong. His PBW ≈ 75 kg, so 540 mL ≈ **7.2 mL/kg**, already **above 6 mL/kg**; it should be **reduced to ~450 mL**.
> - **e:** Hb 11.5 doesn't need transfusion.
> - **a:** inhaled NO improves oxygenation transiently with **no mortality benefit**.
> - **b: prone positioning** is the right next escalation if P/F stays < 150 after PEEP/ventilator optimisation (≥ 12–16 h/day), so some keys accept it. The immediate adjustment on a PEEP of only 5 is **↑ PEEP**.

Q3. A 22-year-old found unresponsive with pinpoint pupils has RR 6, ABG pH 7.21, $PaCO_2$ 70, PaO_2 58, normal A–a gradient. Type and first step? A. Type 1; high-flow O_2 · **B. Type 2 (hypoventilation); bag–mask ventilation + naloxone** · C. ARDS; prone · D. Type 1; CPAP

> **Answer: B** (slides 10, 16).
> - Opioid CNS depression → **pure hypoventilation** (normal A–a gradient).
> - Support ventilation (rescue breaths), **naloxone**; intubate if needed.

Q4. Which finding best distinguishes ARDS from cardiogenic pulmonary oedema? A. Bilateral infiltrates · B. Hypoxaemia · **C. Normal pulmonary capillary wedge pressure** · D. Crackles · E. Tachypnoea

> **Answer: C** (slide 21). ARDS = **non-cardiogenic** oedema with **normal PCWP** (and normal BNP); the others occur in both.

Q5. A patient's SpO_2 stays at 85% despite 100% O_2; ABG PaO_2 is 95 mmHg; blood looks chocolate-brown. He recently received topical benzocaine. Diagnosis and treatment? A. Shunt; PEEP · B. ARDS; prone · **C. Methaemoglobinaemia; methylene blue** · D. Peripheral cyanosis; warming

> **Answer: C** (slide 28 lists methaemoglobinaemia under central cyanosis; *[added]* details).
> - A **saturation gap** (normal PaO_2, low SpO_2) after an oxidising drug.
> - **Methylene blue** (avoid in G6PD deficiency).

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Type 1 failure has high CO_2	❌ Type 1 = **normal/low CO_2**

Trap	Correct thinking
Hypercapnia cut-off 45 vs 50	⚠️ Slide 5 says **> 45**, slide 6 **> 50**; define with pH
Give 15 L O_2 to every hypoxic COPD patient	❌ Target **88–92%**; risk of worsening CO_2
Shunt hypoxaemia corrects with more O_2	❌ **Refractory**; needs PEEP/recruitment
A drowsy hypercapnic patient breathing slower is "settling"	❌ **Tiring**; prepare for ventilation
ARDS = any bilateral infiltrates with hypoxaemia	❌ Must be **not cardiogenic**, **within 7 days** of a risk factor
High BNP + bilateral infiltrates = ARDS	❌ Points to **cardiogenic** oedema
ARDS diagnosed by $PaO_2 < 60$	❌ Severity is by **P/F ratio** with PEEP ≥ 5 (or HFNO)
Large tidal volumes to improve oxygenation	❌ **6 mL/kg PBW**, plateau ≤ 30 (actual body weight is wrong)
Liberal fluids in ARDS	❌ **Conservative fluids / diuretics** once shock resolves
Increase FiO_2 alone in severe ARDS	❌ ↑ **PEEP**, prone, NMB, ECMO
Inhaled NO improves survival	❌ Transient oxygenation only
ARDS shunt is left-to-right	❌ **Right-to-left** (perfused, unventilated alveoli)
Surfactant ↑ in ARDS	❌ ↓ (type II pneumocyte damage)
Cyanosis in severe anaemia	⚠️ May be **absent** (needs > 5 g/dL deoxyHb)
Normal PaO_2 excludes cyanosis	⚠️ **Methaemoglobinaemia**: saturation gap
Slide 24 "Berlin criteria"	⚠️ It shows the **2023 Global Definition** (HFNO, S/F, ultrasound added)